

Physics, 2017 EC, entrance exam, version 3

Grade 12 · Physics · 2017 E.C.

Time allowed: 120 minutes

Questions: 51

Total marks: 51

Name _____ Grade _____ ID No. _____ Date _____

1. Given the vectors $A = (6, -4)$ and $B = (-3, 8)$, in xy -plane both taken from the origin as the reference point. What is the direction of the resultant vector R relative to the positive x -axis? (1)
 - A. 45°
 - B. 53°
 - C. 135°
 - D. 315°
2. Which method involves placing two vectors tail-to-tail and drawing a diagonal to find the resultant? (1)
 - A. Parallelogram law
 - B. Triangle law
 - C. Polygon law
 - D. Scalar addition
3. When comparing the geometric and analytic methods of finding the resultant of two vectors, which of the following is TRUE? (1)
 - A. Both methods always give different magnitudes but the same direction.
 - B. The geometric method gives exact values, while the analytic method gives only estimations.
 - C. Both methods yield the same resultant vector in magnitude and direction when applied correctly.
 - D. The analytical method is only valid when vectors are perpendicular.
4. A vector $A = 2\hat{i} + 3\hat{j} + 4\hat{k}$ and a vector $B = 4\hat{i} + 6\hat{j} + 8\hat{k}$ are given. Which of the following statements is TRUE about the vectors? (1)
 - A. A and B are perpendicular
 - B. A and B are parallel
 - C. The magnitude of A is equal to the magnitude of B
 - D. The dot product of A and B is zero
5. A motorcycle moves with an initial velocity of 15 m/s and accelerates uniformly at a rate of 3 m/s². Which of the following is the best description of the velocity after 6 seconds? (1)
 - A. 33 m/s
 - B. 42 m/s
 - C. 18 m/s
 - D. 45 m/s
6. A particle moves along a straight path with a velocity described by the equation $v(t) = 4t + 3$ m/s where t is the time in seconds. What is the instantaneous acceleration at $t = 5$ seconds? (1)
 - A. 23 m/s²
 - B. 20 m/s²
 - C. 15 m/s²
 - D. 4 m/s²
7. Suppose you are designing a projectile motion system that takes into account both vertical and horizontal displacements. If you wish to minimize the time of flight for the projectile while maintaining a constant range, what variable would you change? (1)
 - A. Decrease the launch angle and adjust the initial velocity to maintain the range.
 - B. Increase the launch angle to achieve greater vertical displacement.
 - C. Change the horizontal velocity to shorten time of flight.
 - D. Increase the initial velocity to reduce the time of flight.

-
8. Two seconds after projection, a projectile is moving at 30° above the horizontal; after one more second it is moving horizontally. The initial speed of the projectile is (1)
- A. $10\sqrt{3}$ m/s
B. 20 m/s
C. 10 m/s
D. $20\sqrt{3}$ m/s
-
9. A flywheel rotates with a uniform angular acceleration. Its angular velocity increases from 20π rad/s to 40π rad/s in 10 seconds. How many rotations did it undergo in this period? (1)
- A. 5
B. 100
C. 150
D. 2
-
10. The equation $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$ represents angular displacement of an object with initial angular velocity ω_0 , undergoing uniform angular acceleration α . Which condition must be met for this equation to be valid? (1)
- A. The object must have a constant angular velocity throughout the motion.
B. The angular acceleration must remain constant during the motion.
C. The initial angular velocity must be zero at the start of the motion.
D. The motion must be linear rather than rotational.
-
11. A wheel having moment of inertia $2 \text{ kg}\cdot\text{m}^2$ about its vertical axis rotates at 60 rpm. The magnitude of torque which can stop the wheel's rotation in one minute would be: (1)
- A. $\pi/12$ Nm
B. $\pi/60$ Nm
C. $\pi/4$ Nm
D. $\pi/15$ Nm
-
12. Two spherical objects, each with mass $5.0 \times 10^5 \text{ kg}$, are placed at a separation of 100 m from each other. What is the gravitational force between them? ($G = 6.7 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$) (1)
- A. $6.7 \times 10^{-7} \text{ N}$
B. $3.0 \times 10^{-4} \text{ N}$
C. $1.7 \times 10^{-3} \text{ N}$
D. $2.0 \times 10^{-1} \text{ N}$
-
13. What does Newton's second law state when it is expressed in terms of momentum? (1)
- A. Force is the time rate of change of momentum.
B. Force is proportional to velocity.
C. Force is equal to the change in acceleration.
D. Force equals mass times velocity.
-
14. A block of mass 50 kg slides over a horizontal distance of 4 m. If the coefficient of friction between their surface is 0.2, then work done against friction is: (1)
- A. 100 J
B. 400 J
C. 200 J
D. 150 J
-
15. Two objects, one with mass 4 kg and the other with mass 8 kg, are moving with the same speed in uniform circular motion on the same horizontal plane, but the radius for the 8 kg object is twice that of the 4 kg object. What is the ratio of centripetal force on the 8 kg object to that on the 4 kg object? (1)
- A. 1:2
B. 2:1
C. 1:4
D. 1:1

-
16. Consider two objects: Object A mass 5 kg, Object B mass 10 kg. If both are subjected to the same applied force, which statement best evaluates the relationship between their accelerations? (1)
- A. Object A will have half the acceleration of Object B
 - B. Object A and Object B will have the same acceleration
 - C. Object A will have double the acceleration of Object B
 - D. The accelerations cannot be compared
-
17. Which statement accurately distinguishes conservative forces from non-conservative forces? (1)
- A. Conservative forces always lead to an increase in mechanical energy.
 - B. Non-conservative forces are path-dependent and can change mechanical energy.
 - C. Conservative forces are always attractive, while non-conservative forces are always repulsive.
 - D. Non-conservative forces never perform work on an object.
-
18. A system uses fixed pulley and movable pulley. If the load is 200 N and the effort force is 100 N and the efficiency is 80%, what is the velocity ratio (VR) of the system? (1)
- A. 2
 - B. 4
 - C. 3.5
 - D. 2.5
-
19. How would you evaluate the effect on the apparent weight of an object when it is immersed in a fluid, compared to its true weight in a vacuum? (1)
- A. The apparent weight will be greater than the true weight due to the buoyant force exerted by the fluid.
 - B. The apparent weight will be the same as the true weight because the fluid has no impact on the force of gravity.
 - C. The apparent weight will be less than the true weight due to the upward buoyant force exerted by the fluid.
 - D. The apparent weight will fluctuate depending on the depth of the fluid.
-
20. Two objects with different temperatures are placed in contact and, after some time, no more heat flows between them. What does this mean? (1)
- A. The cold object took all the heat
 - B. The objects have reached thermal equilibrium
 - C. The objects have different abilities to conduct heat
 - D. Heat is no longer necessary
-
21. A student compares three substances P, Q, and R that undergo heating from solid to gas. Their heating curves show: P has the steepest slope in the solid phase; Q has the longest melting horizontal line; R has the shortest boiling horizontal line. Which conclusion can be drawn? (1)
- A. P has the lowest specific heat capacity in the solid phase.
 - B. Q has the lowest latent heat of fusion.
 - C. R has the highest latent heat of vaporization.
 - D. P has the highest specific heat capacity in the solid phase.
-
22. A student uses 418 J of heat energy to increase the temperature of a 200 g sample of water. If the specific heat capacity of water is $4.18 \text{ J/g}^\circ\text{C}$, by how many degrees Celsius will the temperature increase? (1)
- A. 2.0°C
 - B. 0.38°C
 - C. 1.5°C
 - D. 0.5°C

23. A group of students want to calculate how much energy is required to melt 100 grams of ice at 0°C without changing its temperature. Which concept must they use? (1)

A. Specific heat capacity of ice
B. Thermal conductivity of water
C. Latent heat of fusion of ice
D. Latent heat of vaporization of water

24. Which of the following is a key difference between mechanical and electromagnetic waves? (1)

A. Mechanical waves can travel in a vacuum, but electromagnetic waves cannot.
B. Electromagnetic waves require a medium to travel, whereas mechanical waves do not.
C. Mechanical waves require a medium to travel, while electromagnetic waves do not.
D. Electromagnetic waves are always longitudinal, while mechanical waves are transverse.

25. An object is placed 4 cm in front of a concave mirror. The center of curvature of the mirror is located 40 cm from the pole. What is the distance between the object and the image formed? (1)

A. 9 cm
B. 1 cm
C. 24 cm
D. 8 cm

26. A light passes from lead glass (refractive index $n_1=1.65$) into air ($n_2=1.00$). What is the critical angle for the light ray at the glass-air interface? (1)

A. 53°
B. 30°
C. 37°
D. 45°

27. Which phenomenon occurs when two waves meet and combine to form a wave with larger amplitude? (1)

A. Diffraction
B. Interference
C. Refraction
D. Reflection

28. A designer wants to maximize the total capacitance using three 10 μF capacitors. Which combination should they choose, and why? (1)

A. All in parallel, because it gives the maximum total capacitance.
B. All in series, because it gives the highest voltage rating.
C. One in series with two in parallel, because it balances both voltage and capacitance.
D. It doesn't matter; all combinations give the same result.

29. Two wires A and B are made of the same material. Wire A has twice the length and half the diameter of wire B. What is the ratio R_A / R_B of their resistances? (1)

A. 1
B. 8
C. 4
D. 16

30. A student says: "If the current flows upward in a vertical wire, the magnetic field lines will form clockwise circles around the wire." What is wrong with this statement? (1)

A. Magnetic field lines do not form circles.
B. The direction of the field is counterclockwise, not clockwise.
C. Magnetic fields do not exist around straight wires.
D. The current direction should be downward for counterclockwise fields.

-
- 31.** You are conducting an experiment to demonstrate the force on a current-carrying conductor in a magnetic field. Which new setup will most effectively demonstrate the force? (1)
- A. Use stronger magnetic field and lower the current.
 - B. Use a weaker magnetic field and increase the length of conductor.
 - C. Use stronger magnetic field, increase current, and keep wire perpendicular to field.
 - D. Keep current constant and place wire at 45° to the field.
-
- 32.** A square of side L meters lies in the xy -plane in a region where the magnetic field is given by $B = B_0(2i + 3j + 4k)$ T. The magnitude of flux passing through the square will be: (1)
- A. $2B_0L^2$ Wb
 - B. $3B_0L^2$ Wb
 - C. $4B_0L^2$ Wb
 - D. $4B_0L$ Wb
-
- 33.** What happens if a permanent magnet is cut into two pieces? (1)
- A. The two pieces lose their magnetic properties.
 - B. One piece becomes north pole, the other becomes a south pole.
 - C. Each piece becomes a new magnet with a north and a south pole.
 - D. The magnet retains only its north pole.
-
- 34.** A current of 3 A flows through a long straight wire. At a distance 0.2 m from the wire, the magnetic field is 4×10^{-7} T. What is the magnetic field at a distance of 0.4 m? (1)
- A. 2×10^{-7} T
 - B. 4×10^{-7} T
 - C. 8×10^{-7} T
 - D. 1×10^{-7} T
-
- 35.** A coil with 200 turns has an area of 0.3 m^2 . The magnetic field strength is 0.8 T. Initially, the field makes an angle of 60° with respect to the normal of the coil's plane. After 5 seconds, the coil is rotated to 90° . What is the induced electromotive force? (1)
- A. 4.8 V
 - B. 3.0 V
 - C. 2.4 V
 - D. 1.6 V
-
- 36.** An ideal transformer has a primary coil with 600 turns and a secondary coil with 150 turns. If the primary voltage is 240 V and secondary current is 10 A, what is the primary current? (1)
- A. 1.5 A
 - B. 4.0 A
 - C. 3.0 A
 - D. 2.5 A
-
- 37.** Which of the following phenomena is best explained by Newtonian mechanics in Astronomy? (1)
- A. The Moon revolves around the Earth
 - B. The Sun's light bends when passing near another star
 - C. Galaxies are moving away from each other
 - D. A star becomes a black hole after it dies
-
- 38.** Which of the following is a branch of Earth science that focuses on the study of solid and liquid matter of the Earth? (1)
- A. Astronomy
 - B. Geology
 - C. Meteorology
 - D. Oceanography

-
- 39.** A company wants to improve the efficiency of its electrical power transmission system. Which branch of physics should electrical engineers apply to reduce energy loss? (1)
- A. Thermodynamics
B. Electricals
C. Nuclear physics
D. Electromagnetism
-
- 40.** When holding a pencil between your fingers and applying force, why does a finger on the sharper side feel more pain than the other? (1)
- A. The pencil is heavier on the flat side
B. The pressure is greater on the sharper side
C. The pressure on both fingers is equal
D. The pressure is lower on the finger that feels pain
-
- 41.** A diver in fresh water experiences a pressure of 600 kPa. If they move up by 10 m, what will be the new pressure? (atmospheric pressure = 100 kPa, $g=10 \text{ m/s}^2$, $\rho=1000 \text{ kg/m}^3$) (1)
- A. 500 kPa
B. 520 kPa
C. 540 kPa
D. 550 kPa
-
- 42.** Two hydraulic presses A and B have identical fluid and input forces, but press A has a larger output piston area than press B. Which is true about the force they produce? (1)
- A. Press B produces a larger output force than press A
B. Both presses produce the same output
C. Press A produces a larger output force than press B
D. Output force depends only on input force, not piston area
-
- 43.** A team of engineers is building a cargo ship. They need to ensure the ship remains floating even when fully loaded. What is the best design change they could make? (1)
- A. Increase the ship's height to make it more visible.
B. Increase the surface area of the ship's bottom to displace more water.
C. Use denser materials in the ship's construction to make it more stable.
D. Reduce the ship's total volume to make it more compact.
-
- 44.** A metal block weighs 50 N in air and 42 N when fully submerged in water. What is the density of the metal? (1)
($g=10 \text{ m/s}^2$, $\rho_{\text{water}}=1000 \text{ kg/m}^3$)
- A. 5000 kg/m^3
B. 2000 kg/m^3
C. 2500 kg/m^3
D. 6250 kg/m^3
-
- 45.** An airplane wing is designed so that air moves faster over the top surface than below. If the speed over the top is 200 m/s and under the wing is 150 m/s, which is true? (1)
- A. Pressure above the wing is greater than below.
B. Pressure below the wing is greater than above.
C. Pressure is equal on both sides.
D. Bernoulli's principle does not apply here.
-
- 46.** Which one of the following is a safety measure when using a gas cylinder? (1)
- A. Use the gas cylinder for long hours
B. Store the gas cylinder in a warm area
C. Keep the gas cylinder in a vertical position
D. Keep the gas knob open all the times

-
- 47.** How does electrical conductivity of a P-type semiconductor differ from that of N-type semiconductor? (1)
- A. P-type conducts via electron flow, N-type via hole movement.
 - B. Both conduct through free electrons, but N-type have more.
 - C. P-type semiconductors do not conduct electricity under normal conditions.
 - D. N-type conducts via electron flow, P-type via hole movement.
-
- 48.** What happens to the potential barrier of a P-N junction diode under forward bias? (1)
- A. It increases
 - B. It decreases
 - C. It remains constant
 - D. It becomes infinite
-
- 49.** Designing a power supply for a sensitive audio amplifier requires smoothed signal in the DC output. Which rectifier configuration would be most appropriate, and what additional component would you incorporate? (1)
- A. Half-wave rectifier with a resistor
 - B. Full-wave rectifier with an ammeter
 - C. Full-wave rectifier with a capacitor
 - D. Half-wave rectifier with an inductor
-
- 50.** A transistor with a common-emitter current gain of 100 has a collector current of 10 mA. What is the base current? (1)
- A. 10 μA
 - B. 0.1 μA
 - C. 1 μA
 - D. 100 μA
-
- 51.** Which of the following is a typical advantage of integrated circuits? (1)
- A. Capability to withstand very high voltage
 - B. Large physical size for better handling
 - C. Ability to operate at high speed
 - D. High individual component replacement ease

— END OF EXAMINATION —